

Data Ingestion Report:

Hugging Face Hub Model Metadata

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April 2026

1 Data Source

- **Source:** Hugging Face Hub model metadata (public API)
- **Format:** JSON Lines, fetched via `huggingface_hub` Python library
- **Size:** 2,815,064 models, 1.7 GB JSONL (229 MB gzipped)
- **Supplementary:** 2,019,739 evaluation results (580 MB JSONL)

1.1 Core Fields

Field	Type	Description
<code>modelId</code>	string	Unique ID (<code>owner/model_name</code>)
<code>author</code>	string	Uploader or organization
<code>pipeline_tag</code>	string	Task type (e.g. <code>text-generation</code>)
<code>library_name</code>	string	Framework (e.g. <code>transformers</code>)
<code>downloads</code>	long	30-day download count
<code>likes</code>	long	Like count
<code>lastModified</code>	string	Last update (ISO 8601)
<code>created_at</code>	string	Creation time (ISO 8601)
<code>tags</code>	array	Tags
<code>card_data.license</code>	string	License from model card
<code>card_data.datasets</code>	array	Training datasets
<code>safetensors.parameters.total</code>	long	Parameter count
<code>config.model_type</code>	string	Architecture (e.g. <code>llama</code>)

2 Ingestion

Two stages:

1. **API Fetch** (Python): `ingest_hf_models.py fetch` downloads all public models via `HfApi.list_models()` with `expand` for `safetensors`, `config`, and `cardData`. Outputs `hf_models.jsonl` and `hf_eval_results.jsonl`.
2. **Spark Ingestion** (Scala): `IngestHfModels` reads JSONL with explicit schema, writes raw Parquet.

3 Data Profiling

Scala class: `osspulse.profiling.ProfileHfModels`

Metric	Value
Total records	2,815,064
Total columns	18
Distinct authors	534,877
Distinct pipeline tags	56
Distinct libraries	1,242
Max downloads	210,971,565
Avg downloads	875.25
Median downloads	0

Table 1: Summary statistics.

pipeline_tag	count
(null)	1,923,354
text-generation	353,584
text-classification	114,299
text-to-image	94,737
reinforcement-learning	71,569
asr	30,245
token-classification	27,009
image-classification	26,669
image-text-to-text	25,916
robotics	17,777

Table 2: Top 10 pipeline tags.

library_name	count
(null)	1,496,382
transformers	834,577
peft	243,357
diffusers	103,384
stable-baselines3	25,995
sentence-transformers	17,515
lerobot	15,644
ml-agents	12,567
mlx	8,754
keras	4,672

Table 3: Top 10 library names.

Column	Null Count	Null %
modelId	0	0.0%
author	0	0.0%
pipeline_tag	1,923,354	68.3%
library_name	1,496,382	53.1%
downloads	0	0.0%
likes	0	0.0%
lastModified	0	0.0%
created_at	0	0.0%
card_data.license	1,889,922	67.1%
card_data.datasets	2,646,443	94.0%

Table 4: Null counts.

Key issues: 68% of models lack a `pipeline_tag`; 53% lack `library_name`; 94% lack training dataset metadata; license formatting is inconsistent.

4 Data Cleaning

Scala class: `osspulse.cleaning.CleanHfModels`. Steps:

1. Filter out private and disabled models.
2. Split `modelId` into `owner` and `model_name`.
3. Normalize `pipeline_tag` and `library_name` (lowercase, null $\rightarrow$ "unknown").
4. Parse timestamps (`lastModified` $\rightarrow$ `last_modified_ts`).
5. Extract and normalize `license` from `card_data` with synonym mapping.
6. Extract `parameter_count` from `safetensors` and `model_type` from `config`.
7. Filter records with null `modelId` or invalid format.
8. Deduplicate by `model_id`.

Total cleaned records: **2,815,064**.

5 Dataset Snippet

5.1 Raw Data Sample (top 5 by downloads)

<code>modelId</code>	<code>pipeline_tag</code>	<code>downloads</code>	<code>likes</code>
s-t/all-MiniLM-L6-v2	sentence-similarity	210,971,565	4,719
Qwen/Qwen3-VL-2B-Instruct	image-text-to-text	98,836,433	372
google-bert/bert-base-uncased	fill-mask	60,494,282	2,632
google/electra-base-disc.		47,967,344	95
s-t/paraphrase-multi-MiniLM	sentence-similarity	36,496,231	1,208

5.2 Cleaned Data Sample (top 5 by downloads)

<code>model_id</code>	<code>pipeline_tag</code>	<code>license</code>	<code>downloads</code>	<code>likes</code>	<code>params</code>
s-t/all-MiniLM-L6-v2	sentence-similarity	apache-2.0	210,971,565	4,719	23M
Qwen/Qwen3-VL-2B-Instruct	image-text-to-text	apache-2.0	98,836,433	372	2.1B
google-bert/bert-base-unc.	fill-mask	apache-2.0	60,494,282	2,632	110M
google/electra-base-disc.	unknown	apache-2.0	47,967,344	95	—
s-t/paraphrase-multi-MiniLM	sentence-similarity	apache-2.0	36,496,231	1,208	118M